HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **system requirements** and traceability matrices for active implantable medical devices, ensuring alignment with user needs and regulatory standards.
- Contributed to system architecture discussions, mapping interfaces and integration points for complex medical device subsystems under senior engineer guidance.
- Assisted in the planning and execution of **system verification and validation** activities, documenting test results and supporting the investigation of technical discrepancies.
- Participated in cross-functional design reviews, providing analytical input on **system characterization** and integration challenges to Electrical, Software, and Mechanical teams.
- Supported risk management processes, including contributing to **hazard analysis** and risk assessments for medical device development in compliance with industry guidelines.
- Developed Python scripts to automate data extraction and reporting for system performance metrics, reducing manual effort by approximately **40%**.
- Containerized Java Spring Boot applications using **Docker** and managed deployments to **AWS EC2** instances, improving environment consistency and reducing setup time by **30%**.

SKILLS

Systems Engineering – System Requirements, User Needs, Requirements Traceability, System Architecture, System Integration, Risk Management, Hazard Analysis, Verification, Validation, Design Controls, Product Development Processes

Programming Languages – Java, Python, SQL, C++

Frameworks & Libraries – Spring Boot, Hibernate, JPA, pandas, NumPy

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Agile, Scrum, Linux, Shell Scripting

PROJECTS

Medical Device System Requirements & Traceability

- Developed comprehensive system requirements documentation and maintained bidirectional traceability from user needs through to V&V; test cases for a new implantable device.
- Collaborated with cross-functional teams including Quality Assurance and Regulatory Affairs to ensure all requirements met stringent medical device development standards.
- Assisted in the creation of system architecture diagrams and interface control documents, defining interactions between hardware and software components.

Automated Data Pipeline for System Performance

- Engineered a Python-based data pipeline to ingest and process system logs from various device components, consolidating performance data into a centralized PostgreSQL database.
- Developed SQL queries and scripts to extract and transform data for reporting on key performance indicators, reducing manual data compilation time by an estimated **35%**.
- Implemented automated checks within the pipeline to flag data quality issues and ensure the integrity of the performance metrics used for system characterization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)